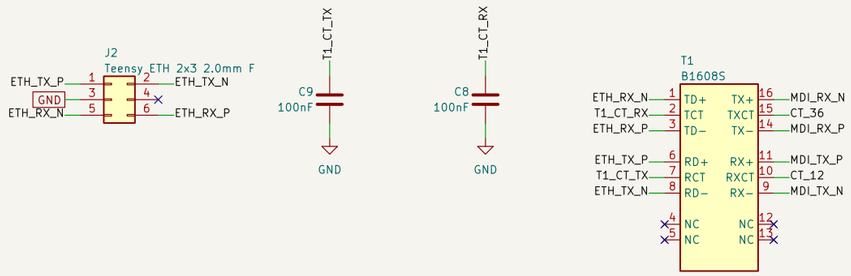
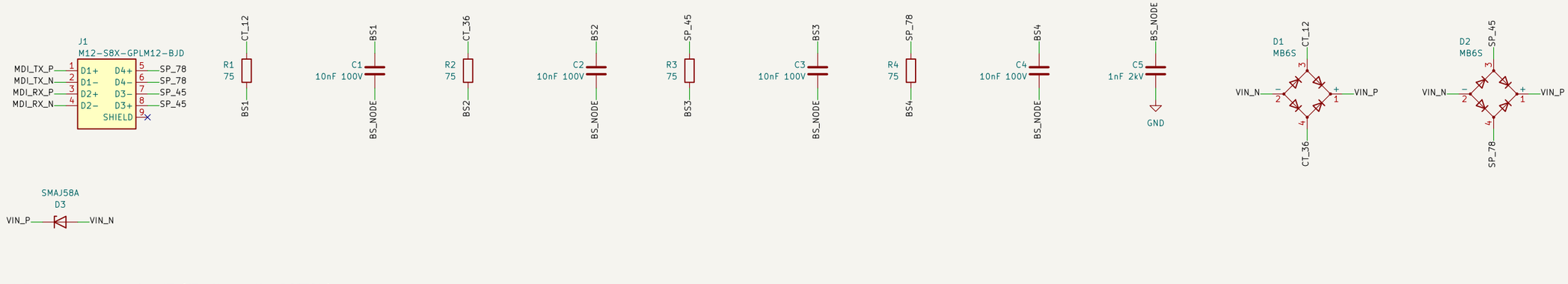
CineTracker PoE board v0.62

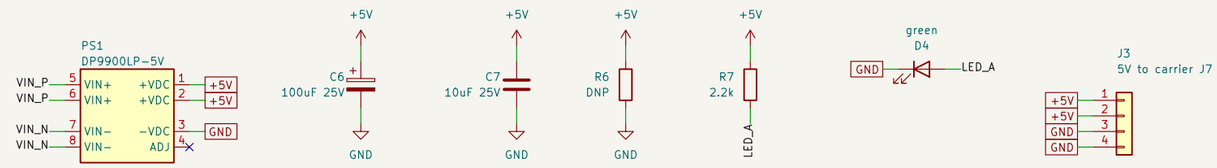
Teensy 4.1 Ethernet link and magnetics (J2 sits on the BOTTOM, on the Teensy's 2x3 2.0 mm header)



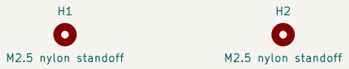
PoE input, PRIMARY side (up to 57 V): M12 X-coded, Bob Smith terminations, bridges, TVS



Isolated 5 V output to carrier J7 (J3 sits on the BOTTOM)



Standoff holes (nylon M2.5; both sit in the isolation band)



Stack: carrier (bottom) / Teensy 4.1 / this board. Coordinates on the PCB are carrier coordinates.

J2 pin order = PJRC Ethernet kit header: 1 TX+, 2 TX−, 3 GND, 4 LED (not used), 5 RX−, 6 RX+ (PJRC teensy41_ethernet_pinout.png).

T1: Teensy TX -> RD winding, Teensy RX -> TD winding (no crossings; DP83825 has auto-MDIX). Dots in phase: 1–16, 6–11.

M12 X-coded pinout (CAZN 8X table) with T568B cordset: 1/2 = RJ45 1/2 (TX), 3/4 = RJ45 3/6 (RX), 8/7 = RJ45 4/5, 5/6 = RJ45 7/8.

Mode A power comes through the T1 cable-side centre taps (D1), Mode B through the spare pairs (D2): any 802.3af/at PSE.

Bob Smith: 75 ohm + 10 nF 100 V per pair, common node -> C5 1 nF 2 kV -> isolated GND (TI SLUA454). J1 pin 9 (shield) is left open.

Isolation: everything on the right of PS1/T1/C5 is PRIMARY. PCB keeps >= 2.6 mm between primary and isolated copper (custom DRC rule).

PS1 needs >= 100 mA load (datasheet). R6 (2512, DNP) is a bleeder footprint in case the measured Teensy + carrier load is lower.

R6 DNP, J3 not assembled by JLC (long-pin header, length set by the stack height), J2 hand-soldered on the bottom side.

Copyright 2026 Dimitris Kontis and CineTracker contributors. CERN-OHL-S-2.0. github.com/dkontisco

Generated from the routed board netlist, then maintained in KiCad

802.3af/at PD on M12 X-coded, isolated 5 V into the carrier, Ethernet into the Teensy 4.1

CineTracker (github.com/dkontiscom/cinetracker)

Sheet: /

File: cinetracker-poe.kicad_sch

Title: CineTracker PoE board

Size: A3	Date: 2026-10-13	**Rev: v0.62**
KiCad E.D.A. 10.0.7		Id: 1/1